

Ziqpu — Entity-Kind Worker Selection

Product Requirements Document: New Feature

Build name: Ziqpu — Entity-Kind Worker Selection

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1. PROBLEM

Ziqpu tells people it can chart “anything you might choose,” but its grounding step asks stock-shaped questions about every entity. A seeker weighing a car, a medication, or an unlisted company is handed “no public signals available” — which reads as we checked and reality is silent, when the truth is we never asked a source that could have answered. The reading then drops silently to its unsourced rung, and the person has no way to tell a real absence from an unasked question. For a product whose entire promise is “measured, and you can check it,” that is not a missing feature; it is a quiet dishonesty.

Supporting Context (optional)

- Measured before this change: a vehicle returned zero grounded signals. Every worker on offer queried SEC endpoints, which structurally cannot describe a car.
- Coverage is ~4,500 chartable US-listed companies out of 7,671 rows. Every non-stock domain — vehicles, drugs, consumer products — is unreachable, not merely sparse.
- The failure is invisible to the user: grounding degrades to the LOCAL · UNSOURCED badge with no explanation, which is the least valuable version of the product.

1a. Opportunity

Asking the right sources per entity turns “chart anything” from a tagline into a shipping capability, and unlocks day-precise origin moments — an FDA approval, a product launch — that the stock universe cannot provide at all.

Market Opportunity

- The addressable set of dated entities grows from ~4,500 US-listed companies to every FDA-approved drug in Drugs@FDA and every VIN-identified vehicle on the road.
- Every source used is U.S. Government public domain (17 USC 105) or CC0, so coverage expands with no licensing cost and no share-alike obligation — the commercial path stays clean.

1b. Users & Needs

Primary user(s): The “curious decider” — someone weighing one specific choice (a car to buy, a medication to start, a company to join or back) who wants a reflective read first and, optionally, a grounded one.

Secondary users: Partners and evaluators running the app on their own machines; the maintainer extending Ziqpu into new domains.

Key User Needs

As a curious decider, I need the app to ground the thing I am actually weighing — not only listed stocks — because my real decisions are rarely about equities.

As a curious decider, I need to know whether “no signals” means nothing exists or nothing was asked, because the product’s promise is that I can check its work.

As a maintainer, I need to add a new domain by adding one worker rather than rewriting the grounding path, because “chart anything” is an open-ended roadmap.

2. PROPOSED SOLUTION

Entity-kind resolution. Before the agent gathers anything, it works out what the entity is from data already in hand — a 17-character VIN, an SEC CIK, or just a name. That kind selects a roster of workers, each one a real public data source, so a vehicle is asked about vehicles and a public filer is asked about filings. Where the kind is genuinely undecidable from the data, the roster carries every plausible worker and the model chooses between them. The grounded reading is then assembled only from what those sources actually returned.

2a. Value Proposition

Curious deciders who struggle with an agent that can only ground US-listed stocks use Ziqpu’s entity-kind resolution to get real, sourced context on the thing they are actually weighing. Unlike the previous fixed roster — which asked SEC endpoints about every entity and returned an unexplained blank for anything else — it routes each entity to sources that can answer, so the app either grounds honestly or says why it cannot.

2b. Top 3 MVP Value Props

The Vitamin (*must-have baseline*): Every entity is asked sources that could plausibly answer it, so a blank result now means a real absence rather than an unasked question.

The Painkiller (*solves the core pain*): A car, a drug, or an unlisted company can be grounded at all — each of them returned nothing whatsoever before this change.

The Steroid (*the magic moment*): A day-precise, citable origin moment for something that is not a company: HUMIRA, approved 2002-12-31 (BLA125057), pulled live and free from the public record.

2c. Goals & Non-Goals

Goals

- A seeker can ground a non-stock entity — a vehicle or a drug — with real sourced signals instead of an empty result.
- Grounding never dispatches a worker that structurally cannot answer for that entity kind.
- A new domain can be added as one worker plus one roster line, without touching the agent loop or the honesty ladder.
- The reading stays impossible to fabricate: every signal shown traces to a source that actually returned it.

Non-Goals

- Per-unit vehicle manufacture dates. The only day-precise record is the door-jamb compliance label — a physical sticker with no API — so we deliberately ship identity and origin place instead of inventing a moment.
- Paid or share-alike data sources (GS1, OpenCorporates, Open Food Facts), excluded to protect the commercial licensing path.
- A user-facing entity-type picker. The kind is inferred from the data; asking the seeker to classify their own choice would push our modelling problem onto them.

2d. Success Metrics

Goal	Signal	Metric	Target
Ground non-stock entities	A vehicle or drug returns real signals	Sourced signals per non-stock entity	≥ 3 for a valid VIN; ≥ 1 dated for a listed drug (achieved: 4 and 3)
Never dispatch an impossible worker	SEC workers are not called for entities with no CIK	SEC dispatches on non-CIK entities	0 — enforced by test
Keep new domains cheap to add	A domain lands without touching the loop	Files changed to add one source	≤ 2 (the worker, plus one roster line)
Never fabricate a signal	Displayed signals all trace to a returning source	Signals shown with no source behind them	0 — the grounding is built from the tool sink, not model prose

3. REQUIREMENTS

User Journey 1: A curious decider grounds a choice that is not a listed stock

Context: This is the journey that simply did not exist before. It matters because the product's stated premise — everything you might choose has a birth moment — is only true if the agent can reach beyond the stock universe.

Sub-journey: Establishing what the entity is

- **[P0]** User can enter a VIN and have the app recognise it as a vehicle without picking a category.
- **[P0]** User can enter a name and have the app work whether or not that name is a listed company.
- **[P0]** User is never asked to classify the entity themselves.
- **[P1]** User can see which sources were consulted, named in the grounded reading.
- **[P2]** User can see that a consulted source returned nothing, distinctly from it not being asked.

Sub-journey: Gathering the real record

- **[P0]** User must approve the grounded pull at the checkpoint before any external call is made.
- **[P0]** User can get real signals for a vehicle: make, model, and the plant where it was assembled.
- **[P0]** User can get a day-precise FDA approval date for a drug, with its application number.

- [P1] User can opt into a broader open-web search as a separate, explicit consent.
- [P2] User can re-run grounding after a source was unavailable.

Sub-journey: Reading the result honestly

- [P0] User sees every signal attributed to the source that returned it.
- [P0] User is never shown a birth date the sources did not supply.
- [P1] User can tell a sourced reading from an unsourced one by its badge.

User Journey 2: A maintainer extends the agent into a new domain

Context: “Chart anything” is open-ended, so the cost of the next domain is the real constraint. This journey exists to keep that cost at one worker.

Sub-journey: Adding a worker

- [P0] Maintainer can add a source by implementing one interface, with no change to the agent loop.
- [P0] Maintainer can attach that worker to an entity kind in a single place.
- [P1] Maintainer can verify the worker against the live source with one ignored test.
- [P2] Maintainer can add a new entity kind without touching existing rosters.

Sub-journey: Proving it stayed honest

- [P0] Maintainer can assert no signal is displayed that a source did not return.
- [P0] Maintainer can assert impossible workers are never dispatched for a kind.
- [P1] Maintainer can assert a failing or absent source degrades rather than fabricating.

4. APPENDIX

Agent-specific sections follow, per the assignment’s note that some agent sections may be new. Everything below describes what is built and pushed on the nightfall branch, not what is planned.

A1. Architecture — the resolution step

Grounding runs as an orchestrator with workers. The orchestrator resolves the entity kind first, selects a worker roster from that kind, then runs the roster. Each worker is one real public source. Two levels decide: code narrows the roster where the answer is determined by data we already hold, and the model chooses within the roster where it is not.

`classify_entity(choice)` → `EntityKind`, in code, no model call: a valid 17-character VIN (digits and uppercase letters, excluding I, O and Q) means `Vehicle`; an SEC Central Index Key means `PublicCompany`; anything else is `Named` — all we know is that it has a name.

`roster_for(kind)` → workers. `PublicCompany` gets `sec_filings`, `sec_financials` and `company_facts`. `Vehicle` gets `vehicle_record` (NHTSA vPIC) and nothing SEC-shaped. `Named` gets `company_facts` and `drug_approval` together, because a bare name could be a private company or a drug and the data cannot separate them — so the model picks.

A2. Tools the agent may call

`sec_filings` — SEC EDGAR submissions: recent filings and SIC industry. Public domain, keyless.

sec_financials — SEC XBRL company facts: latest revenue and total assets, each labelled with the period it covers so a year-to-date total cannot read as annual. Public domain.

company_facts — Wikidata: founding year and headcount. CC0. Guarded by a label-overlap check and an instance-of check so a mis-resolved entity contributes nothing.

vehicle_record — NHTSA vPIC: make, model, model year, assembly plant. Public domain. Returns identity and origin place only; a VIN carries no build date.

drug_approval — openFDA Drugs@FDA: original approval date (day-precise), dosage form, sponsor. Public domain. Exact brand-name search returns NOT_FOUND for non-drugs.

web_news_search — GDELT news. Gated behind a second, explicit human consent at the checkpoint, because the open web is broader and noisier than the structured record.

A3. Blast radius and the human checkpoint

Every worker reaches the public internet, so the whole roster sits behind the existing human-in-the-loop checkpoint: no external call happens without an approval token. The open-web worker requires a second, separate opt-in on top of that. Reads are the only operation — nothing is written, purchased, or sent anywhere on the user's behalf, and no credential is required by any source in the roster.

The residual risk is a wrong answer rather than a harmful action. It is contained two ways: a mis-resolved lookup contributes nothing instead of a wrong fact (the guards in A2), and if the model's roster run collects nothing at all, grounding falls back to the deterministic multi-source fan rather than to an invented answer.

A4. System prompt (the research orchestrator)

"You are a research assistant gathering REAL public signals about an entity for a grounded briefing. Call the tools that fit this entity to collect facts. Do NOT invent, estimate, or infer any data; only report what a tool returns. Call each relevant tool at most once. When you have gathered the available signals, reply with the single word DONE."

The prompt is deliberately narrow: the model selects and calls, it does not narrate. Its final prose is discarded entirely — the grounded reading is assembled from a sink that only tool results are written into, so the model is structurally unable to author a fact.

A5. Evaluations

Roster is a function of the input — a vehicle's roster contains vehicle_record and no SEC worker; a public company's contains all three; a bare name gets both candidates. Asserted on names, no network. PASS.

LIVE DISPATCH — the claim that only exists at inference time: a real tool-calling model, handed two different entities, calls different workers because it is offered different rosters. Run against Qwen3-14B Q6_K served on CUDA (RTX 5080) with --jinja, 32 seconds:

VIN 1HGCM82633A004352 → NHTSA vPIC — "2003 Honda Accord EX-V6", built by American Honda Motor Co., assembled at Marysville, Ohio. No SEC worker was reached.

MANH → SEC EDGAR + Wikipedia · SEC financials (XBRL) · Wikidata — recent filings, industry, revenue \$297.8M for the quarter ending 2026-06-30, founded 1990. No vehicle worker was reached.

Asserted in both directions (a car never reaches SEC, a company never reaches vPIC) and that the two source sets differ at all. PASS — the architecture is proven against real inference rather than mocks.

A panicking worker cannot wedge the app — a tool that unwinds is caught at the tool boundary and becomes an error result the model routes around, and the grounding thread catches anything below it. Without this the checkpoint, which has no cancel, would spin forever. PASS.

Consent names what it spends — the checkpoint prompt is derived from the entity kind, so a vehicle's consent names vPIC and never claims SEC. One shared function renders it for both the desktop app and the MCP server, and a test fails the build if a second copy reappears. PASS.

A car grounds with real signals — live against NHTSA vPIC: “2003 Honda Accord EX-V6”, built by American Honda Motor Co., assembled at Marysville, Ohio. This entity returned nothing before the change. PASS.

A drug grounds with a day-precise moment — live against openFDA: HUMIRA, FDA approval 2002-12-31 (BLA125057). PASS.

A non-drug contributes nothing — “Manhattan Associates Inc” against openFDA returns no items rather than a wrong match. PASS.

No fabricated birth moment — no vehicle signal may contain “born”, “birth” or “built on”, and the standalone year must label itself a model year. PASS.

Honesty spine intact — the grounding is built from the tool sink, a source that contributed nothing is not named, and an all-empty result is marked unsourced rather than grounded. PASS.

A6. Status and links

Shipped to the nightfall branch. Entity-kind selection and the vehicle worker (968d492); the openFDA worker and the ambiguous-kind roster (4b4c1e4); panic containment and kind-aware consent (fa5cb15); the MCP server moved onto the same multi-source path and the same consent function (0c5ca15); the live dispatch proof (c861cb0). 119 agent, 16 UI and 7 MCP tests green, clippy clean.

Every requirement in section 3 is implemented, and every evaluation in A5 passes — including the one that needed real inference. Nothing in this document is aspirational.

Reproduce the dispatch proof: serve a tool-capable model with “cargo run -p model -- serve --port 1235”, then run “cargo test -p agents research -- --ignored --nocapture live_dispatch”.

Repository: github.com/nessaisling-lab/Ziqpu-L2-Cycle-3 — pattern write-up at docs/architecture-pattern.md.

Still open, and deliberately so: a Wikidata P577 worker for consumer-product launch dates, which extends the Named roster without changing the architecture this document describes.